



# Modelling Financial Markets

## Seminar 11: Continuous-Time Models — Hands-on



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## Acronyms used in this material

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BSS: Brownian SemiStationary (process)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ FRED: Federal Reserve Economic Data
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ NLLS: Non-Linear Least Squares
- ▣ OLS: Ordinary Least Squares
- ▣ OU: Ornstein–Uhlenbeck (process)
- ▣ PDE: Partial Differential Equation
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ RFSV: Rough Fractional Stochastic Volatility (model)
- ▣ US: United States
- ▣ VIX: CBOE Volatility IndeX
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

# Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

## Seminar Objectives

### By the end of this seminar you can

- Change measure with Girsanov, price bonds with Feynman–Kac and derive conditional moments of CIR and Heston
- Map between continuous-time models and their discrete-time versions (AR(1), Poisson mixtures)
- Estimate GBM, Vasicek, Merton and Heston parameters with standard errors
- Recognise when standard inference fails: biased  $\hat{\kappa}$ , boundary likelihood-ratio tests, unbounded likelihoods, noisy variance proxies

### Structure

- **Part A:** nine derivations (A1–A9)
- **Part B:** nine computational exercises with inference (B1–B9)
- **Part C:** an open question, a seed for the team project (C1), the critique of an AI answer (C2) and a project seed from the lecture case study on rough and persistent volatility (C3)

- Background: Lecture 11; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [Borak, Härdle & López-Cabrera \(2013\)](#)

## How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
  - ▶ Every Part B exercise ends with interpretation questions
- Time in years, daily step  $\Delta t = 1/252$  (Bitcoin 1/365); log returns; drifts and volatilities per year
- Report every estimate with its standard error or confidence interval (CI)

## Solved and Proposed Problems

### Solved: worked examples in this deck

- Part A: A1 (Girsanov for GBM), A3 (Vasicek bond by Feynman–Kac), A5 (CIR moments and Feller), A7 (Merton cumulants and the information on  $\lambda$ ), A9 (AR(1) to OU with the delta method)
- Part B: B1 (orders of convergence), B2 (GBM on the S&P 500), B4 (Vasicek and the bias of  $\hat{\kappa}$ ), B6 (Merton and a bootstrap LR test)

### Proposed: you solve them

- Part A: A2 (Siegel's paradox and the numéraire; model: A1), A4 (Itô integrals; model: A3), A6 (VIX<sup>2</sup> under Heston; model: A5), A8 (rare large jumps; model: A7)
- Part B: B3 (GBM on BET and Bitcoin; model: B2), B5 (OU for the VIX; model: B4), B7 (jumps in Bitcoin; model: B6), B8 (Heston from the VIX; model: B4, B6), B9 (nonparametric diffusion; model: B4)
- Part C: C1, open question and project seed; C2, critique of an AI answer; C3, is SPY volatility rough and persistent? (lecture case study; model: B1, B4)

## Setup: Python Environment and Data

-  **Open in Colab:** `chapter11_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python  $\geq 3.10$  with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: S&P 500 (1990–2026), BET (2000–2026), Bitcoin (2014–2026), VIX (1990–2026); daily closes up to 18 September 2026
- 3-month US Treasury bill rate (FRED series DTB3), 1954–2026
- Equity indices: weekdays only, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin: 7 days a week

## Formulas Used Today

- ▣ Itô's lemma:  $dX = a dt + b dW \Rightarrow df(X) = (af' + \frac{1}{2}b^2f'')dt + bf' dW$
- ▣ Girsanov:  $dQ/dP = e^{-\theta W_T - \theta^2 T/2}$ ,  $\tilde{W}_t = W_t + \theta t$ ; Feynman–Kac:  $u_t + au_x + \frac{1}{2}b^2u_{xx} - ru = 0$
- ▣ GBM:  $\ln(S_T/S_0) \sim N((\mu - \sigma^2/2)T, \sigma^2 T)$ ;  $\mathbb{E}[S_T] = S_0 e^{\mu T}$
- ▣ OU:  $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$ ,  $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$ ; AR(1):  $b = e^{-\kappa \Delta t}$
- ▣ Merton cumulants:  $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t(\mu_J^2 + \sigma_J^2)$ ,  $\kappa_3 = \lambda \Delta t(\mu_J^3 + 3\mu_J \sigma_J^2)$ ,  $\kappa_4 = \lambda \Delta t(\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$
- ▣ Heston:  $dv = \kappa(\theta - v)dt + \xi\sqrt{v}dW^{(2)}$ ,  $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$ ; Feller:  $2\kappa\theta \geq \xi^2$



## A1: GBM under $P$ and $Q$ (Girsanov) [Solved]

### Task

- ▣ Context: a stock follows GBM  $dS = \mu S dt + \sigma S dW$  under the real-world measure  $P$ , with  $\mu = 8\%$  and  $\sigma = 20\%$  per year **Girsanov (1960)**
- ▣ Assumptions: constant risk-free rate  $r = 3\%$  per year, no dividends, horizon  $T = 10$  years; the exercise uses given numbers, no market data
- ▣ (a) Find the market price of risk  $\theta$  for which the discounted price  $e^{-rt}S_t$  has zero drift, and write the Radon–Nikodym density  $Z_T = dQ/dP$  on  $\mathcal{F}_T$
- ▣ (b) Compute  $\mathbb{E}^P[Z_T S_T]$  in closed form, show that it equals  $S_0 e^{rT}$ , and state whether  $\sigma$  changes under  $Q$
- ▣ (c) Compute  $P(S_{10} < S_0)$  under  $P$  and under  $Q$  from the lognormal law of  $S_T$
- ▣ Report:  $\theta$ ,  $Z_T$ ,  $\mathbb{E}^Q[S_{10}]/S_0$  and the two probabilities; interpretation: which probability is a forecast and which one is a pricing weight?

## A1: Solution (1/2) [Solved]

## Solution

- Step 1: by Itô's product rule,  $d(e^{-rt}S_t) = e^{-rt}S_t[(\mu - r)dt + \sigma dW_t]$
- Step 2: substitute  $dW = d\tilde{W} - \theta dt$ ; the drift becomes  $\mu - r - \sigma\theta$ , which is zero for  $\theta = (\mu - r)/\sigma = (0.08 - 0.03)/0.20 = 0.25$
- Step 3: Girsanov with a constant  $\theta$  gives  $Z_T = \exp(-\theta W_T - \frac{1}{2}\theta^2 T)$ ; the Novikov condition holds because  $\theta$  is constant
- Step 4:  $\mathbb{E}^P[Z_T S_T] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2)T} \mathbb{E}[e^{(\sigma - \theta)W_T}] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2 + (\sigma - \theta)^2/2)T} = S_0 e^{(\mu - \sigma\theta)T} = S_0 e^{rT}$
- Result:  $\mathbb{E}^Q[S_{10}]/S_0 = e^{0.3} = 1.350$ , the growth of a risk-free deposit

## A1: Solution (2/2) [Solved]

## Solution

- Step 5: under  $Q$ ,  $dS = rS dt + \sigma S d\tilde{W}$ :  $\sigma$  is unchanged, because the quadratic variation  $[\ln S]_T = \sigma^2 T$  is a property of the path, the same under equivalent measures
- Step 6:  $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$  with the log drift  $m_P = \mu - \sigma^2/2 = 6\%$  under  $P$  and  $m_Q = r - \sigma^2/2 = 1\%$  under  $Q$
- Under  $P$ :  $\Phi(-0.06\sqrt{10}/0.20) = \Phi(-0.949) = 17.1\%$ ; under  $Q$ :  $\Phi(-0.01\sqrt{10}/0.20) = \Phi(-0.158) = 43.7\%$
- Interpretation: the  $P$ -probability is the forecast of a loss; the  $Q$ -probability is a pricing weight that embeds the risk premium, so it overstates bad outcomes; the whole gap comes from the drift, since  $\sigma$  is common to both

## A2: Siegel's Paradox and the Numéraire [Proposed]

Solution discussed in class

### Task

- Context: the EUR/RON rate  $X_t$  (RON per euro) follows GBM with  $\mu = 2\%$  and  $\sigma = 4.4\%$  per year under  $P$ ;  $Y_t = 1/X_t$  is the RON/EUR rate
- Assumptions: constant interest rates  $r_{RON}$  and  $r_{EUR}$ ; given numbers, no market data
- (a) Apply Itô's lemma to  $f(x) = 1/x$  and write  $dY_t$
- (b) Compute the expected percentage drift of  $Y$  to four decimal places and compare it with  $-\mu = -2\%$
- (c) Compute the log drifts of  $X$  and  $Y$  and compare them
- (d) Show that the forward  $F = X_0 e^{(r_{RON} - r_{EUR})T}$  equals  $\mathbb{E}^{Q^{RON}}[X_T]$ , that  $1/F = \mathbb{E}^{Q^{EUR}}[1/X_T]$ , and write the density  $dQ^{EUR}/dQ^{RON}$
- Model: A1
- Report: the drifts in (b)–(c) and the density in (d); interpretation: both a RON and a euro investor seem to expect a gain from the forward; why is there no arbitrage?

## A3: The Vasicek Bond by Feynman–Kac [Solved]

### Task

- ▣ Context: under the risk-neutral measure  $Q$  the short rate is Vasicek,  $dr = \kappa(\theta - r)dt + \sigma d\tilde{W}$ , with  $\kappa = 0.5$ ,  $\theta = 3\%$ ,  $\sigma = 1\%$  per year and today  $r_0 = 6\%$  [Vasicek \(1977\)](#)
- ▣ Assumptions: the market price of risk is already contained in  $\theta$ ; a zero-coupon bond pays 1 at  $T$ ; given numbers, no market data
- ▣ (a) Use the Feynman–Kac theorem with payoff  $g = 1$  and discount rate  $r$  to write the PDE and the terminal condition for  $P(t, T) = \mathbb{E}^Q[e^{-\int_t^T r_s ds} \mid r_t]$  [Kac \(1949\)](#)
- ▣ (b) Insert the ansatz  $P = \exp(A(\tau) - B(\tau)r)$ ,  $\tau = T - t$ , collect the terms in  $r$  and solve the two ordinary differential equations with  $A(0) = B(0) = 0$
- ▣ (c) Compute  $B$ ,  $A$  and the yield  $y(\tau) = (Br_0 - A)/\tau$  for  $\tau = 1, 5, 10$  years, and the limit  $y_\infty$
- ▣ Report:  $B(5)$ ,  $A(5)$ , the three yields and  $y_\infty$ ; interpretation: why is the yield curve downward-sloping, and why is  $y_\infty$  below  $\theta$ ?

## A3: Solution (1/2) [Solved]

## Solution

- (a) Feynman–Kac:  $P_t + \kappa(\theta - r)P_r + \frac{1}{2}\sigma^2 P_{rr} - rP = 0$  with  $P(T, T) = 1$
- (b) With  $P = e^{A - Br}$ :  $P_t = (-A' + B'r)P$ ,  $P_r = -BP$ ,  $P_{rr} = B^2P$ ; dividing by  $P$ :  
 $-A' + B'r - \kappa(\theta - r)B + \frac{1}{2}\sigma^2 B^2 - r = 0$  for every  $r$
- Terms in  $r$ :  $B' = 1 - \kappa B$ ,  $B(0) = 0$ , so  $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$
- Constant terms:  $A' = -\kappa\theta B + \frac{1}{2}\sigma^2 B^2$ ,  $A(0) = 0$ ; integrating,  $A(\tau) = \left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B - \tau) - \frac{\sigma^2 B^2}{4\kappa}$

## A3: Solution (2/2) [Solved]

## Solution

- (c)  $B(5) = (1 - e^{-2 \cdot 5})/0.5 = 1.836$ ;  $A(5) = (0.03 - 0.0002)(1.836 - 5) - 0.0001 \cdot 1.836^2/2 = -0.0945$
- Yields:  $y(1) = 5.36\%$ ,  $y(5) = 4.09\%$ ,  $y(10) = 3.58\%$ ; limit  $y_\infty = \theta - \sigma^2/(2\kappa^2) = 0.03 - 0.0002 = 2.98\%$
- Interpretation:  $r_0 = 6\%$  is above  $\theta = 3\%$ , so expected future rates fall and the curve slopes down
- $y_\infty < \theta$  because of convexity:  $\mathbb{E}^Q[e^{-\int r}] > e^{-\mathbb{E}^Q[\int r]}$  (Jensen), so uncertainty about rates raises bond prices and lowers long yields by  $\sigma^2/(2\kappa^2)$

## A4: Itô Integrals [Proposed]

Solution discussed in class

### Task

- Context:  $W$  is a standard Brownian motion on  $[0, T]$ ; no data
- (a) Apply Itô's lemma to  $f(x) = x^2$  and show that  $\int_0^T W \, dW = \frac{1}{2}(W_T^2 - T)$
- (b) For  $T = 1$  compute the mean and the variance of this integral, and check the variance with the Itô isometry
- (c) The Stratonovich integral equals  $\frac{1}{2} W_T^2$ : compute its mean for  $T = 1$
- (d) Apply Itô's lemma to  $W_t^3$  and deduce the mean and the variance of  $\int_0^1 W^2 \, dW$
- Model: A3 (Itô's lemma applied to a function of the state)
- Report: the means and variances in (b)–(d); interpretation: why must an Itô integral with a square-integrable integrand have mean zero while the Stratonovich integral does not?



## A5: CIR: Conditional Moments and the Feller Condition [Solved]

### Task

- ▣ Context: the short rate follows CIR,  $dr = \kappa(\theta - r)dt + \sigma\sqrt{r}dW$ , with  $\kappa = 0.11$ ,  $\theta = 5.5\%$ ,  $\sigma = 0.056$  per year, close to the exact CIR fit on daily DTB3, 1954–2007 (B4 (d)) [Cox, Ingersoll & Ross \(1985\)](#)
- ▣ Assumptions: today  $r_0 = 3\%$ ; horizon  $t = 1$  year
- ▣ (a) Apply Itô's lemma to  $e^{\kappa t}r_t$ , take expectations and derive  $\mathbb{E}[r_t | r_0]$
- ▣ (b) Apply Itô's lemma to  $r_t^2$ , write the ordinary differential equation for  $m_2(t) = \mathbb{E}[r_t^2]$ , solve it and derive  $\text{Var}(r_t | r_0)$
- ▣ (c) State the transition law  $2c r_t | r_0 \sim \chi_\nu^2(\lambda)$  and compute  $c$ ,  $\nu$ ,  $\lambda$  and the 5% and 95% quantiles of  $r_1$
- ▣ (d) Show that the Feller condition  $2\kappa\theta \geq \sigma^2$  is equivalent to  $\nu \geq 2$  and evaluate the Feller ratio
- ▣ Report: mean, standard deviation,  $c$ ,  $\nu$ ,  $\lambda$ , the 90% interval and the Feller ratio; interpretation: why is the interval asymmetric around the mean, and can the rate become negative?

## A5: Solution (1/2) [Solved]

## Solution

- (a)  $d(e^{\kappa t}r) = e^{\kappa t}(\kappa r dt + dr) = \kappa\theta e^{\kappa t} dt + \sigma e^{\kappa t}\sqrt{r} dW$ ; the integrand is square-integrable ( $\mathbb{E}[r_s] < \infty$ ), so the stochastic integral has mean zero and  $\mathbb{E}[r_t] = \theta + (r_0 - \theta)e^{-\kappa t}$
- (b)  $d(r^2) = 2r dr + \sigma^2 r dt = [(2\kappa\theta + \sigma^2)r - 2\kappa r^2] dt + 2\sigma r^{3/2} dW$
- Taking expectations:  $m'_2 = (2\kappa\theta + \sigma^2)\mathbb{E}[r_t] - 2\kappa m_2$ ; solving and subtracting  $\mathbb{E}[r_t]^2$ :  

$$\text{Var} = r_0 \frac{\sigma^2}{\kappa} (e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta \sigma^2}{2\kappa} (1 - e^{-\kappa t})^2$$
- Numbers:  $\mathbb{E}[r_1] = 0.055 + (0.03 - 0.055)e^{-0.11} = 3.26\%$ ; standard deviation 0.94%

## A5: Solution (2/2) [Solved]

## Solution

- (c)  $c = \frac{2\kappa}{\sigma^2(1-e^{-\kappa t})} = 673$ ,  $\nu = \frac{4\kappa\theta}{\sigma^2} = 7.72$ ,  $\lambda = 2c r_0 e^{-\kappa t} = 36.2$
- Quantiles of  $\chi^2_\nu(\lambda)$  divided by  $2c$ : 90% interval for  $r_1$  is [1.85%, 4.92%]
- (d)  $\nu = 2 \cdot (2\kappa\theta/\sigma^2)$ , so  $2\kappa\theta \geq \sigma^2 \Leftrightarrow \nu \geq 2$ ; Feller ratio  $2\kappa\theta/\sigma^2 = 3.86$
- Interpretation: the noncentral  $\chi^2$  law is skewed to the right, because volatility  $\sigma\sqrt{r}$  grows with the level: the interval reaches further above the mean than below it
- With  $\nu \geq 2$  zero is never reached, so the rate stays strictly positive, unlike Vasicek; if Feller fails, zero is attainable but negative values never are: the 8 negative DTB3 quotes after 2008 lie outside CIR

## A6: The VIX under Heston [Proposed]

Solution discussed in class

### Task

- ▣ Context: under  $Q$  the Heston variance follows  $dv = \kappa^Q(\theta^Q - v)dt + \xi\sqrt{v}d\tilde{W}^{(2)}$ ; the squared VIX is  $y_t = \text{VIX}_t^2/100^2 = \frac{1}{\tau}\mathbb{E}^Q\left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t\right]$ ,  $\tau = 30/365$
- ▣ Given numbers:  $\kappa^Q = 5$ ,  $\theta^Q = 0.044$ ; the lecture CIR regression on  $y_t$  gives  $\hat{\xi} = 0.56$
- ▣ (a) Using A5 (a) under  $Q$ , write  $\mathbb{E}^Q[v_s \mid v_t]$ , integrate over  $[t, t + \tau]$  and show that  $y_t = a + b v_t$ ; give  $a$  and  $b$
- ▣ (b) Compute  $\kappa^Q\tau$ ,  $b$  and  $a$  for the given numbers
- ▣ (c) From  $dy = b dv$ , express the diffusion of  $y$  through  $y$ , relate the estimated  $\xi_y$  to  $\xi$  at  $v_t = \theta^Q$  and compute the corrected  $\xi$
- ▣ (d) State which parameters  $(\kappa, \theta, \rho)$  the regression on  $y_t$  recovers, and under which measure
- ▣ Model: A5
- ▣ Report:  $a$ ,  $b$  and the corrected  $\xi$ ; interpretation: is  $\hat{\xi} = 0.56$  too high or too low, and what else is needed to estimate  $\kappa^Q$ ?

## A7: Merton: Cumulants and the Information on $\lambda$ [Solved]

### Task

- ▣ Context: Merton model with  $\sigma = 15\%$ ,  $\lambda = 5$  jumps per year,  $\mu_J = -3\%$ ,  $\sigma_J = 4\%$ , daily step  $\Delta t = 1/252$  Merton (1976)
- ▣ For (c): the S&P 500 sample spans  $T = 36.7$  years; the lecture MLE is  $\hat{\lambda} = 101.9$  with s.e. 10.5
- ▣ (a) Write the cumulant generating function  $K(u)$  of the daily log return, derive  $\kappa_k = \lambda \Delta t \mathbb{E}[Y^k]$  for  $k \geq 3$ , and compute  $\kappa_2$  split into its diffusion and jump parts
- ▣ (b) Compute  $\kappa_3$ ,  $\kappa_4$ , the skewness  $\kappa_3/\kappa_2^{3/2}$  and the excess kurtosis  $\kappa_4/\kappa_2^2$
- ▣ (c) Assume every jump is observed: write the Poisson log-likelihood of  $N_T$ , derive the Fisher information for  $\lambda$  and the s.e. of  $\hat{\lambda} = N_T/T$  at  $\lambda = 5$  and at  $\lambda = 101.9$
- ▣ Report:  $\kappa_2$  to  $\kappa_4$ , skewness, excess kurtosis and the two standard errors; interpretation: why is the MLE s.e. much larger than the s.e. with observed jumps?

## A7: Solution (1/2) [Solved]

## Solution

- (a)  $K(u) = u m \Delta t + \frac{1}{2} u^2 \sigma^2 \Delta t + \lambda \Delta t (\mathbb{E}[e^{uY}] - 1)$ ; the  $k$ -th derivative at 0 gives  $\kappa_k = \lambda \Delta t \mathbb{E}[Y^k]$  for  $k \geq 3$
- $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2) = 0.893 \times 10^{-4} + 0.496 \times 10^{-4} = 1.389 \times 10^{-4}$ ; jumps carry 35.7% of the variance
- (b)  $\mathbb{E}[Y^3] = \mu_J^3 + 3\mu_J \sigma_J^2$ ,  $\mathbb{E}[Y^4] = \mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4$ :  $\kappa_3 = -3.393 \times 10^{-6}$ ,  $\kappa_4 = 3.399 \times 10^{-7}$
- Skewness  $-2.07$ , excess kurtosis  $17.6$ : five modest jumps a year make daily returns strongly non-Normal

## A7: Solution (2/2) [Solved]

## Solution

- (c)  $\ell(\lambda) = N_T \ln \lambda - \lambda T + \text{const}$ ;  $\ell'' = -N_T/\lambda^2$ , so  $I(\lambda) = \mathbb{E}[N_T]/\lambda^2 = T/\lambda$  and  $\text{s.e.}(\hat{\lambda}) = \sqrt{\lambda/T}$
- At  $\lambda = 5$ ,  $T = 36.7$ :  $\text{s.e.} \sqrt{5/36.7} = 0.37$ , i.e. 7.4% of  $\lambda$
- At  $\hat{\lambda} = 101.9$ , observed jumps would give  $\text{s.e.} 1.67$ ; the MLE  $\text{s.e.}$  is 6.3 times larger
- Interpretation: the jumps are latent; the Poisson mixture must separate them from diffusive moves, which destroys most of the information about  $\lambda$

## A8: Rare, Large Jumps [Proposed]

Solution discussed in class

### Task

- ▣ Context: Merton model with  $\sigma = 15\%$  but rare crashes:  $\lambda = 0.5$  per year,  $\mu_J = -10\%$ ,  $\sigma_J = 5\%$ , daily step  $\Delta t = 1/252$ ; sample span  $T = 36.7$  years
- ▣ (a) Compute the daily skewness and excess kurtosis with the cumulant formulas of A7
- ▣ (b) Compute the probability of observing no jump in 1 year and in 10 years
- ▣ (c) Even with every crash observed, compute the s.e. of  $\hat{\lambda}$  over 36.7 years and the number of years needed for a 20% relative s.e.
- ▣ (d) Explain what the lecture MLE ( $\hat{\lambda} \approx 100$ ) captures instead
- ▣ Model: A7
- ▣ Report: skewness, excess kurtosis, the two probabilities, the s.e. and the years; interpretation: can crash risk be estimated from the history of one index?



## A9: From an AR(1) to an OU Process, with Standard Errors [Solved]

### Task

- ▣ Context: an AR(1) estimated on  $n = 600$  months (50 years) of a rate in decimals, step  $\Delta = 1/12$ :  
 $x_{t+1} = 0.0012 + 0.97 x_t + \varepsilon_t$ , residual s.d.  $\hat{s}_\varepsilon = 0.0025$
- ▣ Assumptions: the rate is an exactly sampled OU process, so  $b = e^{-\kappa\Delta}$ ,  $a = \theta(1 - b)$ ,  $s^2 = \sigma^2(1 - b^2)/(2\kappa)$
- ▣ (a) Invert the three relations to obtain  $\kappa$ ,  $\theta$ ,  $\sigma$  and the half-life  $\ln 2/\kappa$
- ▣ (b) Use  $\text{Var}(\hat{b}) \approx (1 - b^2)/n$  and the delta method to compute the s.e. of  $\hat{\kappa}$  and of the half-life
- ▣ (c) Keep  $T = 50$  years but sample daily ( $\Delta = 1/252$ ): compute  $b$ , the s.e. of  $\hat{\kappa}$  and the limit  $\sqrt{2\kappa/T}$
- ▣ Report:  $\kappa$ ,  $\theta$ ,  $\sigma$ , half-life with s.e. and the daily s.e.; interpretation: does daily sampling make  $\kappa$  more precise?

## A9: Solution (1/2) [Solved]

## Solution

- (a)  $\kappa = -\ln b/\Delta = -12 \ln 0.97 = 0.366$ ;  $\theta = a/(1 - b) = 0.0012/0.03 = 4.0\%$
- $\sigma = s\sqrt{2\kappa/(1 - b^2)} = 0.0025\sqrt{2 \cdot 0.366/(1 - 0.97^2)} = 0.88\%$ ; half-life  $\ln 2/0.366 = 1.90$  years
- (b)  $\text{s.e.}(\hat{b}) = \sqrt{(1 - 0.97^2)/600} = 0.0099$ ;  $\partial\kappa/\partial b = -1/(b\Delta)$ , so  $\text{s.e.}(\hat{\kappa}) = 0.0099/(0.97/12) = 0.123$
- Half-life: derivative  $-\ln 2/\kappa^2$ , so  $\text{s.e.} = (\ln 2/\kappa^2) \text{s.e.}(\hat{\kappa}) = 0.64$  years: a 95% CI of roughly  $\pm 1.2$  years

## A9: Solution (2/2) [Solved]

## Solution

- (c) Daily:  $b = e^{-\kappa/252} = 0.99855$  and  $n = 12\,600$ ; the same formulas give s.e. 0.121
- In general  $\text{Var}(\hat{\kappa}) \approx (1 - b^2)/(nb^2\Delta^2) = (e^{2\kappa\Delta} - 1)/(T\Delta)$ ; as  $\Delta \rightarrow 0$  it tends to  $2\kappa/T$ ; limit  $\sqrt{2\kappa/T} = 0.121$
- Interpretation: 21 times more observations give the same precision:  $\kappa$  is a long-span parameter, identified by calendar time, not by the number of points (lecture, Section 7)

## B1: Orders of Convergence [Solved]

### Task

- ▣ Setting: test equation  $dX = 2X dt + X dW$ ,  $X_0 = 1$ ,  $T = 1$ , with exact solution  $X_T = \exp(1.5 + W_T)$  [Higham \(2001\)](#); 20,000 Brownian paths on a fine grid of  $2^{11}$  steps, coarsened to  $\Delta t = 2^{-11}, \dots, 2^{-5}$  (common random numbers)
- ▣ (a) For each  $\Delta t$  compute the strong error  $\mathbb{E}|X_T - X_T^{\Delta t}|$  of Euler–Maruyama and Milstein, averaged over the 20,000 paths
- ▣ (b) Regress  $\ln(\text{error})$  on  $\ln \Delta t$  by OLS; since all step sizes use the same paths, obtain the s.e. and a 95% CI of each slope by resampling whole paths (1,000 bootstrap samples, all step sizes together)
- ▣ (c) Compute the weak error  $|\mathbb{E}[X_T^{\Delta t}] - \mathbb{E}[X_T]|$  of Euler–Maruyama exactly, from  $\mathbb{E}[X_T^{\Delta t}] = (1 + 2\Delta t)^{1/\Delta t}$ , and by Monte Carlo; estimate both slopes
- ▣ Report: a table of the four slopes with bootstrap s.e. and CI (none for the deterministic exact-mean slope)

### Interpretation

- ▣ Are the slopes consistent with the theoretical orders 1/2 and 1? Why does the Monte Carlo weak error give a wrong slope, and how many paths would resolve it?

## B1: Results [Solved]

|                                   | Slope | s.e.  | 95% CI         |
|-----------------------------------|-------|-------|----------------|
| Strong, Euler–Maruyama            | 0.518 | 0.003 | [0.512, 0.525] |
| Strong, Milstein                  | 0.986 | 0.001 | [0.984, 0.987] |
| Weak, Euler–Maruyama, exact mean  | 0.985 | –     | deterministic  |
| Weak, Euler–Maruyama, Monte Carlo | 0.511 | 0.269 | [0.247, 1.289] |

- Step 1: strong slopes are close to 0.5 and 1; halving  $\Delta t$  halves the Milstein error but cuts the Euler error only by  $1 - 2^{-1/2} = 29\%$
- Step 2: the path-bootstrap CIs measure Monte Carlo noise only; they can exclude 1/2 and 1 because the errors at coarse steps are pre-asymptotic: the local slopes between the two finest steps are 0.492 (Euler) and 0.998 (Milstein)

 MFM\_ch11\_convergence

## B1: Reading the Weak Error [Solved]

- Step 3: the exact weak slope, a deterministic number with no sampling CI, is close to 1, as theory predicts (local slope at the finest steps 0.998)
- Step 4: the s.d. of  $X_T$  is  $e^2\sqrt{e-1} = 9.69$ , so the Monte Carlo s.e. of the mean is  $9.69/\sqrt{20\,000} = 0.068$
- Step 5: at  $\Delta t = 2^{-11}$  the weak error is 0.0072, ten times smaller than this s.e., so the Monte Carlo curve flattens and its slope is wrong
- Step 6: to resolve it at the 95% level,  $(1.96 \cdot 9.69/0.0072)^2 \approx 6\,937\,381$  paths are needed
- Interpretation: for expected values the Monte Carlo sample size, not the time step, is usually the binding constraint

 MFM\_ch11\_convergence

## B2: GBM on the S&P 500 [Solved]

### Task

- Data: S&P 500 daily closes from `data/market/GSPC.INDX.csv`, 2 January 1990 – 18 September 2026, weekdays without stale quotes; log returns  $r_t$ ,  $\Delta t = 1/252$ ,  $n = 9\,245$ ,  $T = 36.7$  years
- (a) Compute the MLE  $\hat{\sigma}^2 = \sum (r_t - \bar{r})^2 / (n\Delta t)$ ,  $\hat{m} = \bar{r} / \Delta t$ ,  $\hat{\mu} = \hat{m} + \hat{\sigma}^2 / 2$ , with  $\text{s.e.}(\hat{\sigma}) = \hat{\sigma} / \sqrt{2n}$ ,  $\text{s.e.}(\hat{m}) = \hat{\sigma} / \sqrt{T}$  and a 95% CI for  $m$
- (b) Test normality with Jarque–Bera and independence with Ljung–Box on  $|r_t|$  (20 lags), both at 5%
- (c) Compute the variance ratio of non-overlapping 20-day sums to 20 times the daily variance (GBM: 1)
- (d) Compute the years of data needed for a 95% CI of  $\pm 1$  percentage point on  $m$
- (e) The s.e. in (a) assume GBM; recompute them by a moving-block bootstrap (blocks of 63 days, 999 samples) [Künsch \(1989\)](#), and give  $\hat{\sigma} \sqrt{(k+2)/(4n)}$ , the s.e. of  $\hat{\sigma}$  for independent returns with excess kurtosis  $k$
- Report: estimates with GBM and block-bootstrap s.e., test statistics with critical values and decisions

### Interpretation

- Which parameter is known well and which badly, and why? Which GBM assumption does each test reject? Can you still use  $\hat{\sigma}$  for option pricing?

## B2: Results [Solved]

- ▣ Step 1 (GBM s.e.):  $\hat{\sigma} = 18.0\%$  (s.e. 0.13),  $\hat{m} = 8.3\%$  (s.e. 3.0),  $\hat{\mu} = 8.3\% + \hat{\sigma}^2/2 = 10.0\%$
- ▣ Step 2: GBM 95% CI for  $m$ : [2.5%, 14.2%];  $\pm 1$  point would need  $(1.96\hat{\sigma}/0.01)^2 = 1\,243$  years
- ▣ Step 3 (e): s.e. of  $\hat{\sigma}$ : 0.34 with kurtosis, 1.15 by block bootstrap, 8.7 times the GBM value;  $m$ : block s.e. 2.5, CI [3.5%, 13.3%]
- ▣ Step 4: Jarque–Bera 45 805 against the 5% critical value 5.99 (excess kurtosis 10.9):  $p < 0.001$ , Normality rejected
- ▣ Step 5: Ljung–Box on  $|r_t|$ : 13 381 against  $\chi^2_{20,0.95} = 31.4$ , lag-1 ACF 0.27:  $p < 0.001$ , independence rejected
- ▣ Step 6: variance ratio 0.75, below 1: mild mean reversion of daily moves
- ▣ Interpretation:  $\sigma$  is estimated far more precisely than the drift even without GBM;  $\hat{\sigma}$  is an **average** of a time-varying volatility [Cont \(2001\)](#): a benchmark for pricing, which needs the risk-neutral dynamics of variance, not a calibration



## B3: GBM on BET and Bitcoin [Proposed]

Solution discussed in class

### Task

- Data: BET daily closes from data/market/BET.csv, 2000–2026, weekdays without stale quotes,  $\Delta t = 1/252$ ; Bitcoin from data/market/BTC-USD.CC.csv, 17 September 2014 – 18 September 2026, 7 days a week,  $\Delta t = 1/365$
- (a) Repeat B2 (a), (d) and (e) for each market:  $\hat{\sigma}$ ,  $\hat{m}$  with GBM and block-bootstrap 95% CIs and the years needed for  $\pm 1$  point
- (b) Repeat B2 (b) and (c): Jarque–Bera, Ljung–Box on  $|r_t|$  with 20 lags at 5%, and the 20-day variance ratio
- Model: B2
- Report: one table with both markets

### Interpretation

- Is the drift of Bitcoin significantly positive? Which market is further from GBM, and in which dimension? What does a variance ratio above 1 for BET suggest?

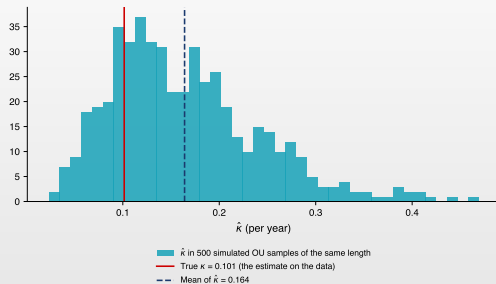
## B4: Vasicek and the Bias of $\hat{\kappa}$ [Solved]

### Task

- Data: 3-month US Treasury bill rate (FRED series DTB3), daily, 4 January 1954 – 18 September 2026, in decimals;  $\Delta t = 1/252$ ,  $n = 18\,170$  days (72.1 years)
- (a) Regress  $x_{t+1}$  on  $x_t$  by OLS (the exact AR(1) representation of the OU process), then recover  $\kappa$ ,  $\theta$ ,  $\sigma$ , compute delta-method s.e., a 95% CI for  $\kappa$  and the half-life Vasicek (1977)
- (b) Simulate 500 exact OU paths of the same length, started from the stationary law, with the estimates as the truth; re-estimate  $\kappa$  on each
- (c) Compute the bias (mean of the 500 estimates minus the truth), compare it with  $4/T$  Tang & Chen (2009), and give the bias-corrected  $\kappa$  and half-life
- (d) Test  $\kappa = 0$  with a simulated null: 500 random walks of the same length from  $x_0$ , residual s.d. as estimated;  $p$  = share of  $\hat{\kappa}$  at or above the observed one
- Report: estimates with s.e., the simulated distribution of  $\hat{\kappa}$  and the corrected values

### Interpretation

- Is mean reversion significant before and after the bias correction? What does the bias do to pricing and hedging with Vasicek?

B4: Simulated Bias of  $\hat{\kappa}$  [Solved]

- 500 exact OU samples of 72.1 years with  $\kappa = 0.101$ : the histogram lies mostly to the right of the truth

## B4: Results [Solved]

- Step 1:  $\hat{b} = 0.99960$ , so  $\hat{\kappa} = -\ln \hat{b} / \Delta t = 0.101$  (s.e. 0.051), 95% CI [0.001, 0.202]
- Step 2:  $\hat{\theta} = 4.55\%$  (s.e. 1.55),  $\hat{\sigma} = 1.33\%$ , half-life  $\ln 2 / \hat{\kappa} = 6.8$  years
- Step 3: simulated mean  $\hat{\kappa} = 0.164$  (s.d. 0.078): bias 0.063, close to  $4/T = 0.055$ ; 78% of the estimates exceed the truth
- Step 4: bias-corrected  $\kappa = 0.101 - 0.063 = 0.039$  (approximate 95% interval  $0.039 \pm 1.96 \cdot 0.078 = [-0.114, 0.192]$ ), half-life 17.8 years instead of 6.8
- Step 5 (d): under a random walk,  $\hat{\kappa}$  has 95% quantile 0.213;  $p = 0.287$  for the observed 0.101
- Interpretation: the Wald CI barely excludes zero and ignores the bias; neither the calibrated test in (d) nor the corrected interval rejects  $\kappa = 0$  at 5%. The corrected half-life is long, yet over 72 years the point estimate would shrink a deviation to 0.06 of its size: the data are compatible with slow reversion, but cannot exclude a random walk. Under a constant market price of risk,  $\kappa^Q = \kappa$ , so the upward bias makes Vasicek understate long-maturity rate risk

## B4 (d): Exact CIR, Euler and CKLS [Solved]

## Task

- Data: DTB3 on 4 January 1954 – 31 December 2007 only (after 2008 the series has 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods, and a near-zero regime), daily ( $n = 13\,487$ ,  $\Delta = 1/252$ ) and month-end values ( $n = 647$ ,  $\Delta = 1/12$ )
- (d1) Maximise the exact CIR likelihood ( $2c\,r_{t+\Delta} \mid r_t \sim \chi^2_\nu(\lambda)$ , A5) and the Euler (Gaussian) pseudo-likelihood; compare  $\hat{\kappa}$  and  $\hat{\theta}$  with Hessian s.e. [Cox, Ingersoll & Ross \(1985\)](#)
- (d2) Fit CKLS,  $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ , by the Euler quasi-likelihood with  $\gamma$  free; compute the Hessian, sandwich and HAC sandwich (Newey–West, Bartlett kernel,  $\lfloor 4(n/100)^{2/9} \rfloor$  lags) s.e. of  $\hat{\gamma}$  [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) [Newey & West \(1987\)](#)
- (d3) Test  $\gamma = 0$  (Vasicek) and  $\gamma = 1/2$  (CIR) by the Gaussian LR and by the HAC Wald test at 5% (critical value  $\chi^2_{1,0.95} = 3.84$ ), at both frequencies
- Report: a table of  $\hat{\kappa}$  (s.e.) and  $\hat{\theta}$  by model and frequency;  $\hat{\gamma}$  with the Hessian, sandwich and HAC s.e.; the four statistics per frequency

## Interpretation

- Which volatility specification do the data prefer, and does the answer change with the sampling frequency? Which test do you trust, LR or HAC Wald?

## B4 (d): Estimates [Solved]

| Model          | Daily                 |                    | Month-end             |                    |
|----------------|-----------------------|--------------------|-----------------------|--------------------|
|                | $\hat{\kappa}$ (s.e.) | $\hat{\theta}$ (%) | $\hat{\kappa}$ (s.e.) | $\hat{\theta}$ (%) |
| Vasicek, exact | 0.158 (0.073)         | 5.39               | 0.201 (0.083)         | 5.36               |
| CIR, exact     | 0.113 (0.060)         | 5.48               | 0.153 (0.070)         | 5.43               |
| CIR, Euler     | 0.093 (0.060)         | 5.55               | 0.144 (0.069)         | 5.45               |
| CKLS, Euler    | 0.080 (0.057)         | 5.84               | 0.144 (0.068)         | 5.46               |

- Step 1: exact and Euler CIR differ by less than one s.e. of  $\hat{\kappa}$  at both frequencies: the discretisation error is small next to the long-span uncertainty
- Step 2: all  $\hat{\kappa}$  lie between 0.08 and 0.20 with s.e. 0.06–0.08: none is estimated precisely, whatever the likelihood

## B4 (d): Tests on $\gamma$ [Solved]

- Step 3:  $\hat{\gamma} = 0.77$  daily (s.e.: Hessian 0.008, sandwich 0.020, HAC 0.033 with 11 lags);  $\hat{\gamma} = 0.60$  monthly (Hessian 0.031, sandwich 0.134, HAC 0.149 with 6 lags)
- Step 4,  $\gamma = 0$ : LR 7926.6 daily and 285.6 monthly; HAC Wald 545.1 and 16.1, all above 3.84: Vasicek is rejected at 5% at both frequencies
- Step 5,  $\gamma = 1/2$ : daily LR 1139.34 and HAC Wald 67.01 reject CIR (22.13 with 55 lags); monthly LR 10.07 rejects, but the HAC Wald  $0.45 < 3.84$  does not
- Interpretation: volatility rises with the level ( $\gamma$  between  $1/2$  and  $1$ ); omitted volatility clustering makes the scores heteroskedastic and serially correlated: the Hessian s.e. are far too small and the LR over-rejects; the contemporaneous sandwich fixes only heteroskedasticity, so trust the HAC Wald test
- Chan, Karolyi, Longstaff & Sanders (1992) found  $\hat{\gamma} \approx 1.5$  on 1964–1989: the elasticity depends on the sample and on how volatility clustering is handled

## B5: Mean Reversion of the VIX [Proposed]

Solution discussed in class

### Task

- Data: VIX daily closes from data/market/VIX.INDX.csv, 1990–2026, weekdays;  $x_t = \ln VIX_t$ ,  $\Delta t = 1/252$ ,  $n = 9\,283$
- (a) Fit the OU process by the exact AR(1) as in B4 (a); convert  $\hat{\kappa}$  into a half-life in trading days,  $252 \ln 2 / \hat{\kappa}$ , with a 95% CI from the CI of  $\kappa$
- (b) Estimate the bias of  $\hat{\kappa}$  with 300 simulated OU paths of the same length
- (c) Compute the sample ACF of  $x_t$  at lags 1, 5, 20, 60, 120 and compare it with the OU prediction  $\hat{b}^k$
- Model: B4
- Report:  $\hat{\kappa}$ , the long-run level  $e^{\hat{\theta}}$ , the half-life with CI, the bias and the ACF table

### Interpretation

- Why is the bias much less important here than for interest rates? What does the ACF pattern say about a one-factor OU model?



## B6: Merton and a Bootstrap Likelihood-Ratio Test [Solved]

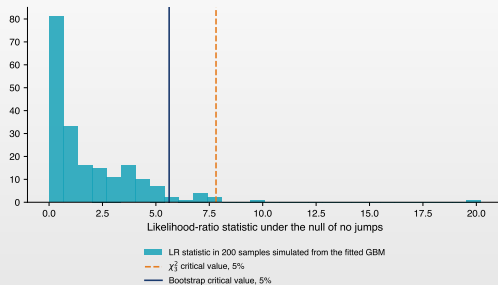
### Task

- Data: S&P 500 daily closes from `data/market/GSPC.INDX.csv`, 2 January 1990 – 18 September 2026, weekdays without stale quotes; log returns  $r_t$ ,  $\Delta t = 1/252$ ,  $n = 9\,245$ ,  $T = 36.7$  years
- (a) Fit Merton ( $m, \sigma, \lambda, \mu_J, \sigma_J$ ) by maximum likelihood on the Poisson mixture truncated at 12 jumps a day, over the bounded space  $\sigma \geq 5\%$  (see (d)), from six starting points, with Hessian s.e. [Merton \(1976\)](#)
- (b) Compute the likelihood-ratio statistic against GBM ( $\lambda = 0$ ) on this restricted space
- (c) Obtain its null distribution by parametric bootstrap: 200 GBM samples of the same length, both models refitted with the same restriction and the same six starts; give the 95% critical value, the number of exceedances and  $p = (1 + \#\{LR^* \geq LR\})/201$
- (d) Compute the profile log-likelihood with  $\sigma$  fixed at 5%, 2%, 1%, the MLE under  $\sigma \geq 5\%$ , and decide whether the likelihood is bounded [Honoré \(1998\)](#)
- Report: parameters with s.e., LR, bootstrap and  $\chi^2_3$  critical values,  $p$ -value, the profile table

### Interpretation

- Why is  $\chi^2_3$  not valid here? Does the fitted model match the kurtosis and skewness of the data? Is the likelihood bounded, and what does the reported MLE then mean?

## B6: Bootstrap Null Distribution [Solved]



- LR statistics in 200 GBM samples, refitted by both models, with the bootstrap and  $\chi^2_3$  critical values at 5%

## B6: Results [Solved]

- Step 1:  $\hat{\sigma} = 9.44\%$  (0.32),  $\hat{\lambda} = 101.9$  (10.5) per year,  $\hat{\mu}_J = -0.162\%$  (0.034),  $\hat{\sigma}_J = 1.44\%$  (0.06)
- Step 2: implied skewness  $-0.31$  (data  $-0.36$ ), excess kurtosis  $3.71$  (data  $10.9$ ): two thirds of the kurtosis is missed
- Step 3:  $LR = 2401$  ( $\sigma \geq 5\%$ , not binding); bootstrap 95% critical value  $5.62$  against  $\chi^2_{3,0.95} = 7.81$ ; 0 of the 200 bootstrap values exceed the LR, so  $\hat{p} = (1 + 0)/201 = 0.0050$  (95% Monte Carlo upper bound  $0.015$ ): GBM is rejected at 5%
- Interpretation: under  $H_0$ ,  $\lambda = 0$  is on the boundary and  $\mu_J$ ,  $\sigma_J$  are unidentified, so Wilks' theorem fails; GBM is rejected, but the test cannot tell jumps from stochastic volatility; the i.i.d. jump model is not the right description of the tails

## B6 (d): Profile Likelihood [Solved]

| $\sigma$ (fixed) | log-likelihood | $\hat{\lambda}$ per year | $\hat{\sigma}_J$ (%) |
|------------------|----------------|--------------------------|----------------------|
| 9.4% (MLE)       | 29 500         | 101.9                    | 1.44                 |
| 5%               | 29 422         | 302                      | 0.93                 |
| 2%               | 29 265         | 555                      | 0.70                 |
| 1%               | 29 165         | 710                      | 0.62                 |

- Step 4: the profile falls as  $\sigma$  shrinks while  $\hat{\lambda}$  grows; the MLE  $\hat{\sigma} = 9.4\%$  satisfies  $\sigma \geq 5\%$ , so the constraint does not bind
- Step 5: yet  $\sup = +\infty$ : set  $m\Delta t$  equal to one observed return; its term  $\ln(e^{-\lambda\Delta t}\varphi(0; 0, \sigma^2\Delta t))$  equals 6.0, 10.7, 19.9, 38.3 at  $\sigma = 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}$ , while the other returns keep finite densities from the jump components
- Interpretation: as for Normal mixtures, the MLE is an interior local maximum; the global LR against GBM would be infinite, so the test in (b)–(c) is defined on  $\sigma \geq 5\%$ , with the same rule for data and bootstrap [Honoré \(1998\)](#)

## B7: Jumps in Bitcoin [Proposed]

Solution discussed in class

### Task

- Data: Bitcoin daily closes from `data/market/BTC-USD.CC.csv`, 17 September 2014 – 18 September 2026, 7 days a week; log returns,  $\Delta t = 1/365$
- (a) Fit Merton by maximum likelihood as in B6 (a); report the parameters with s.e., the implied excess kurtosis and the jump share of variance
- (b) Apply the Lee–Mykland test at  $\alpha = 1\%$  with a bipower window of  $K = 16$  days [Lee & Mykland \(2008\)](#); count the jumps per year and their mean absolute size
- Model: B6
- Report: the Merton estimates and the Lee–Mykland counts side by side

### Interpretation

- Why do  $\hat{\lambda}$  and the Lee–Mykland count differ by two orders of magnitude? Which estimate would you use to price a one-week Bitcoin option?

## B8: Heston Parameters from the VIX [Proposed]

Solution discussed in class

### Task

- Data: S&P 500 (as in B2) and VIX (data/market/VIX.INDX.csv) on common days, 1990–2026;  $y_t = (\text{VIX}_t/100)^2$  in place of  $v_t$ ,  $\Delta t = 1/252$  [Heston \(1993\)](#)
- (a) Run the discretised CIR regression  $(y_{t+\Delta} - y_t)/\sqrt{y_t} = \kappa\theta\Delta t/\sqrt{y_t} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$ ; give  $\kappa$ ,  $\theta$ ,  $\xi$ , the OLS s.e. and a 95% CI for  $\kappa$  (model-based: valid only for homoskedastic, serially uncorrelated  $\varepsilon_t$ ; add HAC s.e.)
- (b) Estimate  $\rho$  as the correlation of return and variance shocks, with a Fisher-z 95% CI (which assumes i.i.d. bivariate Normal shocks); compute the Feller ratio  $2\kappa\theta/\xi^2$
- (c) Repeat (a)–(b) on 1990–2007 and on 2008–2026
- Model: B4, B6
- Report: a table with the full sample and the two halves

### Interpretation

- Why is  $\sqrt{\theta}$  from the VIX above the realised volatility? Are the parameters stable, and is the Feller condition met in both halves?

## B8 (d)–(e): The Affine Map and Realised Variance [Proposed]

Solution discussed in class

### Task

- Data: as in B8, plus SPY 5-minute realised variance including the overnight return (Chapter 9, Quantlets/Ch\_09/ch9\_rv\_spy.csv), 12 October 2020 – 18 September 2026, annualised with  $\Delta t = 1/252$
- (d) Correct  $\hat{\xi}$  for the affine map  $y = a + b v$  of A6 with  $\kappa^Q = 5$  and state which estimates need no correction
- (e) On the common days, estimate  $\rho$  twice, once with  $y_t$  and once with realised variance in place of  $v_t$ , each with a Fisher-z 95% CI
- Model: A6, B8
- Report: the corrected  $\xi$  and the two correlations with CIs

### Interpretation

- Why is  $\rho$  from realised variance much closer to zero than  $\rho$  from the VIX, and which one would you use in a Heston model?

## B9: Nonparametric Diffusion of the T-Bill Rate [Proposed]

Solution discussed in class

### Task

- Data: DTB3 daily, 4 January 1954 – 31 December 2007,  $\Delta = 1/252$ ; the parametric fits of B4 (d) [Bandi & Phillips \(2003\)](#)
- (a) On a grid of 60 levels between the 2% and 98% quantiles of  $r$ , compute Nadaraya–Watson estimates of  $\mathbb{E}[\Delta r \mid r]/\Delta = a(r) + O(\Delta)$  and  $\mathbb{E}[(\Delta r)^2 \mid r]/\Delta = b^2(r) + O(\Delta)$  (drift and diffusion as  $\Delta \rightarrow 0$ ) with a Gaussian kernel and  $h = 1.06 s_r n^{-1/5}$ , plus 95% heteroskedasticity-robust pointwise bands
- (b) Overlay the Vasicek, CIR and CKLS diffusions of B4 (d) and compute the share of levels at which each lies inside the band
- (c) Repeat with  $h/2$  and  $2h$ ; also report the log-slope of  $\hat{b}(r)$  and the share of levels where the drift band contains zero
- Model: B4
- Report: the chart of the lecture (Section 7) and a table by bandwidth

### Interpretation

- Which parametric model lies inside the band, and where does each fail? Can the drift be distinguished from zero? Does the answer depend on  $h$ , and what do the bands ignore?



## C1: Which Continuous-Time Model Fits BET and Bitcoin? [Proposed]

Solution discussed in class

### Task

- Data: BET (data/market/BET.csv, 2000–2026,  $\Delta t = 1/252$ ) and Bitcoin (data/market/BTC-USD.CC.csv, 2014–2026,  $\Delta t = 1/365$ ) daily log returns; no implied-volatility index covering the full sample is available for these markets
- (a) Fit GBM (B2), Merton (B6) and Heston by the CIR regression of B8 with the proxy  $RV_t = \frac{1}{21\Delta t} \sum_{j=0}^{20} r_{t-j}^2$  in place of  $v_t$
- (b) Simulate 100 samples of the same length from each fitted model and compute excess kurtosis, the Hill statistic of the largest losses ( $k = 5\%$  of all returns), skewness and the sum of the ACF of  $|r_t|$  over lags 1–20
- (c) For each model and statistic, report the percentile  $q$  of the data value in the simulated distribution: a descriptive check (parameters from the same data, not refitted on the simulated samples)
- (d) Open extension, with no reference solution: fit Bates (Heston plus Merton jumps) for latent volatility (simulated moments, particle filter) and repeat (b)–(c)
- Report: medians with 90% ranges and the table of percentiles  $q$

### Interpretation

- Which features does each model reproduce, and which none? Is Bitcoin a distinct class [Pele et al. \(2023\)](#)?

## C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

### The student asks an AI assistant

- "Assume the S&P 500 follows a GBM with  $\mu = 8\%$  and  $\sigma = 20\%$  per year. What is the probability that it ends the year below today's level? Check it by simulation."

### The AI answer: text

- GBM:  $dS = \mu S dt + \sigma S dW$
- By the chain rule,  $d \ln S = dS/S = \mu dt + \sigma dW$ , so  $\ln(S_T/S_0) \sim N(\mu T, \sigma^2 T)$
- $P(S_T < S_0) = \Phi(-\mu\sqrt{T}/\sigma) = \Phi(-0.40) = 0.345$ , with  $\Phi$  the distribution function of the standard Normal distribution
- $\mathbb{E}[S_T] = S_0 e^{\mu T}$ : on average the index ends the year 8.3% higher
- The simulation in the code gives 0.345 and confirms the formula



## C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
mu, sigma, T, n, N = 0.08, 0.20, 1.0, 252, 20_000
dt = T / n
rng = np.random.default_rng(1)
Z = rng.standard_normal((N, n))
lnS = np.cumsum((mu - 0.5*sigma**2)*dt + sigma*dt*Z,
                axis=1)
print((lnS[:, -1] < 0).mean())
```

## C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

### Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
  - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

### Hints

- ▣ Apply Itô's lemma to  $f(S) = \ln S$  (A1): which term does the ordinary chain rule miss?
- ▣ Brownian increments over  $\Delta t$  have variance  $\Delta t$ : what is their standard deviation?
- ▣ Run the code: what does it actually print?

## C3: Is SPY Volatility Rough and Persistent? [Proposed]

Solution discussed in class

### Task

- Context: case study [Bennedsen, Lunde & Pakkanen \(2022\)](#), Sections 3, 5.1 and 6; model: B1 (log-log slope), B4 (OLS autoregression); conventions for (a)–(d): next slide
- Data: SPY 5-minute bars (data/market/intraday/SPY.US\_5m.csv, 2020–2026), 09:30–16:00 New York time
- (a) For  $\Delta = 30$  minutes, 65 minutes and 1 day, compute  $\hat{\sigma}_t^2 =$  bipower variation [Barndorff-Nielsen & Shephard \(2004\)](#) of the 5-minute log returns in the bin, divided by  $\Delta$  and, intraday, by the flexible Fourier seasonal factor [Andersen & Bollerslev \(1997\)](#)
- (b) Estimate  $\alpha$  by OLS on the log-variogram with  $k = 1, \dots, 6$  (eq. (3.1)) and by NLLS averaged over  $m = 10, \dots, 20$  (Sections 3.1 and 4.1); report 90% moving-block bootstrap CIs [Künsch \(1989\)](#)
- (c) Given  $\hat{\alpha}$ , fit  $\beta$  (Cauchy) and  $\lambda$  (Gamma-BSS) to the first  $\lceil n^{1/3} \rceil$  autocorrelations of  $\log \hat{\sigma}_t$ , without and with the measurement-noise correction (Section 3.2)
- (d) For  $\Delta = 65$  minutes, forecast the variance summed over the next  $h = 1, 5, 20$  steps with log-HAR (eq. (6.1)), RFSV [Gatheral, Jaisson & Rosenbaum \(2018\)](#) (Section 6) and the Cauchy and Gamma-BSS conditional means with correction (6.2), on a rolling window of 200 observations; compute MSE, QLIKE and the 90% MCS [Hansen, Lunde & Nason \(2011\)](#) (blocks of 6 steps, Table 4)
- Report:  $\hat{\alpha}$  with CIs,  $\hat{\beta}^*$ ,  $\hat{\lambda}^*$  (Table 3, Panel A), MSE, QLIKE, MCS (Table 5, Panel A), code and a two-page note

### Interpretation

- Is SPY log-volatility rough ( $\alpha < 0$ ) and long-memory ( $\beta < 1$ )? Why do OLS and NLLS differ at 30 minutes? Do models that incorporate persistence forecast better than RFSV and log-HAR?

## C3: Conventions [Proposed]

- An illustrative application, not an exact replication: 5-minute bars instead of every trade, so no pre-averaging (Appendix A)
  - ▶ (a) Complete 09:30–16:00 sessions only (78 bars); first return of a session:  $\log(\text{close}/\text{open})$  of its first bar, then close-to-close; no overnight return
  - ▶ A bin of  $M$  returns ( $M = 6, 13, 78$ ):  $BV = \frac{\pi}{2} \frac{M}{M-1} \sum_{j=2}^M |r_j| |r_{j-1}|$ ,  $\hat{\sigma}_t^2 = BV_t/\Delta$ ,  $\Delta$  in trading days (30 minutes = 1/13)
  - ▶ Seasonal factor: flexible Fourier form for the volatility  $\sigma^s$  of each bin, scaled so that  $(\sigma^s)^2$  averages 1 over the session; divide  $\hat{\sigma}_t^2$  by  $(\sigma_t^s)^2$
- (b) Moving-block bootstrap of  $\log \hat{\sigma}_t$ : blocks of  $\lceil n^{1/3} \rceil$  steps, 999 resamples, variogram only from pairs inside one block (joins create artificial jumps),  $\alpha$  refitted each time, percentile CIs
  - ▶ Under long memory the block bootstrap has no general coverage guarantee: report these CIs as exploratory
  - ▶ (c) Lags  $h$  and  $\lambda$  are in steps of length  $\Delta$ : state the unit of every  $\hat{\lambda}$  and compare with Table 3 only in the same unit
- (d) Model  $x_t = \log \hat{\sigma}_t^2$  (de-seasonalised, de-meanned); level forecast  $\exp(m + s^2/2)$  (eq. (6.2)) for Cauchy and Gamma-BSS, the same correction with the OLS residual variance for log-HAR, the correction of [Gatheral, Jaisson & Rosenbaum \(2018\)](#) (Section 5.2) for RFSV
  - ▶ Forecast  $\sum_{k=1}^h \widehat{\sigma}_{t+k\Delta|t}^2 (\sigma_{t+k\Delta}^s)^2 \Delta$  against  $\sum_{k=1}^h BV_{t+k\Delta}$  (eq. (6.3)); out of sample: the last two years, as in Section 6.1; parameters and seasonal factor from data up to  $t$  only

## Common Pitfalls

- ▣ Forgetting the  $-\sigma^2/2$  term: confusing the drift of the price with the drift of the log price
- ▣ Reporting a drift estimate without its standard error  $\sigma/\sqrt{T}$
- ▣ Mixing units: annual parameters with daily  $\Delta t$ , trading days with calendar days (Bitcoin)
- ▣ Reading  $\hat{\kappa}$  from a short sample as fast mean reversion: it is biased upwards
- ▣ Using  $\chi^2$  critical values for tests with a parameter on the boundary
- ▣ Treating  $(\text{VIX}/100)^2$  as  $v_t$ : it is affine in  $v_t$  with risk-neutral parameters, so  $\xi$  is attenuated and the level contains a risk premium
- ▣ Hessian standard errors from a Gaussian quasi-likelihood: use sandwich standard errors, HAC when the scores are serially correlated

## Take-Home Assignment (Team Project)

- Choose one asset not covered in class (a BVB stock, a currency or a crypto-asset)
- Fit GBM, Merton and a stochastic-volatility model; report every parameter with a standard error or a bootstrap CI
- Test jumps with Lee–Mykland and compare the jump count with the Merton intensity
- Simulate each model and rank them by how well they reproduce four stylised facts
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`



## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Applied Computational Finance
- ▣  XFG Advanced Methods in Quantitative Finance

### Code and data

- ▣ Seminar notebook (Colab); Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_11](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_11)

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